Establishing a Baseline: Public Reception, Understanding, and Responses to Flooding Forecasts and Warnings in the Contiguous United States

Reference Report

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This report describes the results of an annual nationwide survey on flooding in the United States. The 2024 Flooding and Society Survey (FL24) was designed by the Institute for Public Policy Research and Analysis (IPPRA) at the University of Oklahoma and the Nurture Nature Center (NNC). It is the inaugural survey in an annual series and follows the same format as IPPRA's Severe Weather and Society Surveys (WX), Tropical Cyclones and Society Surveys (TC), and Winter Weather and Society Surveys (WW). FL24 was fielded June 12 – June 18, 2024, using an online questionnaire that was completed by 1,200 U.S. adults (age 18+) who were recruited from an Internet panel that matches the characteristics of the U.S. population as estimated in the U.S. Census. Following the format of IPPRA's other surveys and incorporating NNC's past work in the flooding domain, the FL24 survey establishes baseline measures of the extent to which U.S. adults receive, understand, and respond to flood forecasts and warnings, while also asking questions that address different types of flooding (including coastal and flash flooding). Questions like these will be asked annually to track changes over time. Other questions specific to FL24 evaluated the understanding and use of a flagship NWS flood product, the Weather Prediction Center's Excessive Rainfall Outlook, and impact-based warning comprehension associated with flooding events.

1. INTRODUCTION

1.1. Background

The mission of the National Weather Service (NWS) is to provide weather, water, and climate data, forecasts, and warnings for the protection of life and property and enhancement of the national economy. Currently, the NWS uses warning verification statistics (such as probability of detection, false alarm ratio, and warning lead time) to measure the extent to which they are achieving this mission. This measurement strategy assumes that increasing the accuracy and

timeliness of forecasts and warnings will generate increases in actions leading to the protection of life and property. While necessary, improvements to forecasts and warnings are not always sufficient to generate this outcome. Rather, there are many social conditions that must be met. For example, members of the public must receive, understand, and respond to the forecasts and warnings that the NWS issues. If one or more of these conditions are not met, improvements to forecast and warning technology will not necessarily advance the mission of the NWS. As such, we recommend that the NWS consider these and other social conditions when measuring performance before, during, and after policy changes. Accomplishing this task requires at least three things:

1. Valid and reliable indicators of social conditions (such as forecast and warning reception, understanding, and response);
2. Baseline measures of these indicators under the current forecast and warning system;
3. A protocol that will allow the NWS to track these indicators over time and space so that program managers can empirically detect changes that occur as, when, and where the NWS implements new policies.

The Flooding and Society Survey (FL Survey, for short), an annual nationwide survey on flooding in the United States, was designed with these tasks in mind. This survey is the first survey in the flooding series. It follows the format of the Severe Weather and Society Survey series that began in 2017 (see Silva et al. 2017), the Tropical Weather and Society Survey series that began in 2020 (see Ripberger et al. 2020), and the Winter Weather and Society Survey series that began in 2021 (see Krocak et al. 2021). Pursuant to item one (above), FL24 measured a variety of social indicators such as forecast reception, comprehension, and response. FL24 also measured public trust in the NWS, extreme weather and climate risk perceptions, interpretations of probabilistic language, and flooding preparedness.

Manuscripts and conference presentations using this data are currently under development. Analysis and results of data from other Extreme Weather and Society surveys, WX, TC, and WW, have been published, and highlights from this research are listed below:

- Bitterman et al. 2023: Numbers or Words?: How Perceptions of Probability Comprehension Influence the Communication Pathway for Weather Forecasting Products [LINK]
- Krocak et al. 2023: Public Information Priorities across Weather Hazards and Time Scales. [LINK]
- Ripberger et al. 2022: There's a Chance of What? Assessing Numeracy Skills of Forecasters, Emergency Managers, and Publics to Improve TC Product Uncertainty Communication, IDSS, and Training. [LINK]
- Lenhardt, E., et al., 2020: How Likely is That Chance of Thunderstorms? A Study of How National Weather Service Forecast Offices Use Words of Estimative Probability and What They Mean to the Public. [LINK]
- Allan, J., et al., 2020: Geographic Distributions of Extreme Weather Risk Perceptions in the United States. [LINK]

- Allan, J., et al., 2020: Probability versus Consequences in Public Perceptions of Tornado Risk. [LINK]
- Wehde, W., et al., 2020: Public Willingness to Pay for Probabilistic Weather Forecasts. [LINK]
- Ernst, S., et al., 2020: Weathering the Effects of Affect: Modeling the Causes and Consequences of Extreme Weather Affective Experience [LINK]
- Krocak, M., et al., 2021: An analysis of tornado warning reception and response across time: leveraging respondent's confidence and a nocturnal tornado climatology. [LINK]
- Ernst, S., et al., 2021: Colorful Language: Investigating Public Interpretation of the Storm Prediction Center Convective Outlook. [LINK]
- Krocak, M., et al., 2021: Exploring the differences in SPC convective outlook interpretation using categorical and numeric information [LINK]
- Ernst, S., et al., 2022: Keeping Context in Probabilities: Comparing Public Responses to Absolute and Relative Probability Forecasts of Tornadoes [LINK]
- Krocak, M., et al., 2022: Forecast Information Type across Time: What do People Want and Need? [LINK]
- Krocak, M., et al., 2023: Public Information Priorities across Weather Hazards and Time Scales [LINK]

This report presents an overview of the FL24 sampling and weighting methodology and a reproduction of the survey instrument with weighted frequencies (or means) for the questions that elicited numeric responses. Before that, we provide a brief overview of survey research via the Internet.

1.2. Survey Research via the Internet

Technological developments and telecommunication trends, such as the declining number of landline phones, the increasing use of cellular phones, and the continuing expansion of high speed Internet services, have made probabilistic (often referred to as “random”) sampling of the U.S. national population for the administration of lengthy surveys on complex issues infeasible for several reasons:

- The total universe of households without phone service of any kind is unknown;
- Wired phone lines are no longer maintained in a sufficient fraction of U.S. households to represent the national population, and members of households that do have land-line phone services differ systematically from households without wired phones;
- The number of households with wired phones that are exclusively used for purposes other than routine phone calls, such as home alarms or medical alert services, is unknown;
- The number of individuals and households having both a wired phone and a cell phone or those having more than one cell phone is unknown;
- The number of households and individuals having access to Internet services suitable for taking web-based surveys is unknown;
- The number of individuals who have access to Internet services from their workplace is unknown, and of those, the number of individuals who can take surveys while at work is unknown;
- Cell phones may be unsuitable for lengthy surveys, especially for respondents who are otherwise occupied, and surveys conducted using cell phones may incur costs to

prospective respondents that might ultimately discourage survey participation or reduce the quality of data collected; and
- Face-to-face interviews or printed postal surveys of the U.S. public require long collection periods, often with low response rates, and are prohibitively expensive for many research projects.

Because of the challenges for probabilistic sampling due to the factors listed above, increasingly, quality academic surveys of the U.S. public on complex subjects, such as extreme weather, are being conducted via the Internet.

With increasing Internet access, the demographics of the online population are becoming more representative of the U.S. population, but samples recruited to participate in Internet surveys cannot be truly random samples of the U.S. public. All surveys, regardless of collection methods, include an element of self-selection bias because, even if a perfectly random sample could be constructed, the final decision to participate must be made voluntarily by each respondent, and thus some degree of self-selection is unavoidable. This means that, even when derived from a theoretically perfect random sample, the demographic characteristics of survey respondents may not exactly reflect U.S. population parameters. Non-probabilistic samples, such as those used to administer surveys of the public via the Internet, involve greater degrees of self-selection because participants first voluntarily agree to enter a pool or stream of U.S. residents willing to take surveys online, and then each member of that group must decide whether to participate in a survey opportunity. This requires the administration of Internet surveys that are as demographically representative as possible, and, as with other survey administration options, it warrants caution in presenting findings as statistically representative of views of the entire adult U.S. population.

2. SAMPLING, DEMOGRAPHICS, AND DATA COLLECTION

The sample of survey participants for FL24 was provided by Verasight, a research and marketing company that maintains a diverse panel of Internet users in the U.S. who have agreed to participate in online surveys. Verasight recruits these panelists in multiple ways, including random address sampling, random person-to-person texting, and dynamic targeting.

For FL24, Verasight used a dynamic sampling process to identify eligible panelists (U.S. adults who live in the contiguous U.S.) and invite them to participate in the survey. To begin, invitations were sent to an anonymous group of panelists that match the demographic characteristics of the target population. As the first group of panelists completed the survey, Verasight sent additional invitations to panelists based on demographic targets. If a given group was underrepresented (relative to U.S. Census estimates), more invitations were sent to that group; if the group was overrepresented, fewer invitations were sent. This resulted in a diverse sample of survey participants that is generally representative of the U.S. adult population. Table

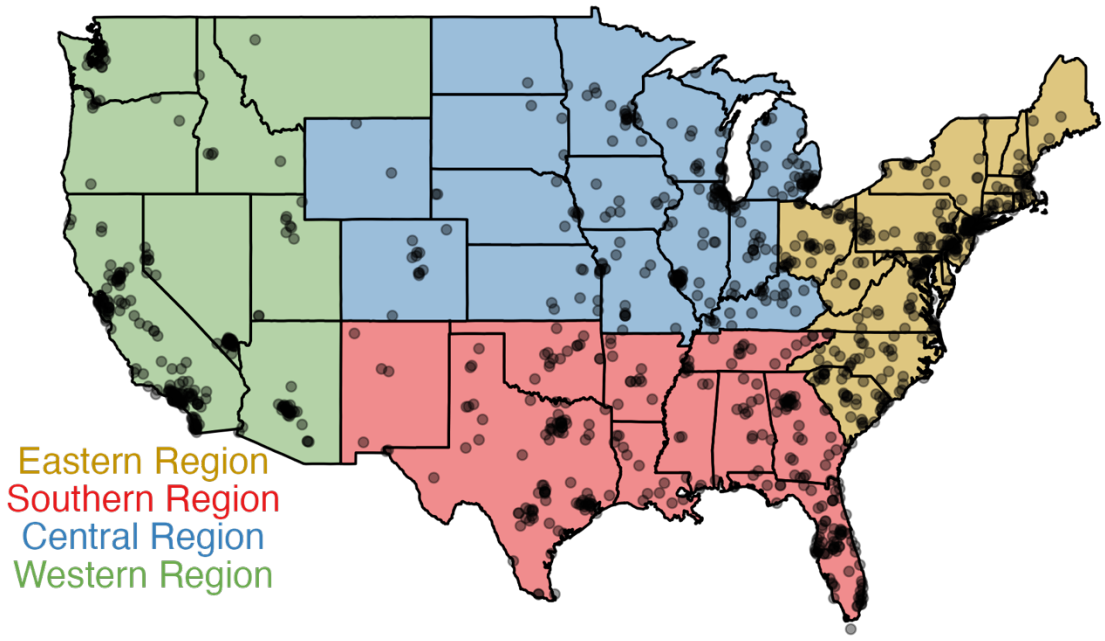
2.1 demonstrates this representativeness by comparing key national and regional population estimates from the U.S. Census to the demographic characteristics of FL24 respondents. Verasight also imposes a series of data quality checks to identify and remove duplicate completions, speeders, non-human responses, and more.

Table 2.1: Demographic Representativeness of FL24 Respondents

	U.S. Adult Population* (%)	Participants (%)
Gender		
Female	50.9	50.0
Male	49.1	50.0
Age		
18 to 24	11.7	6.5
25 to 34	17.6	15.2
35 to 44	16.8	19.2
45 to 54	15.8	16.0
55 to 64	16.6	15.6
65 and up	21.6	27.4
Ethnicity		
Hispanic	83.0	86.5
Non-Hispanic	17.0	13.5
Race		
White	77.4	76.8
Black or African American	13.1	12.2
Asian	6.1	4.9
Other Race	3.4	6.1
NWS Region		
Eastern	31.7	29.9
Southern	27.3	24.9
Central	20.5	20.6
Western	20.5	24.6
*Population estimates were obtained from the U.S. Census Annual Estimates of the Resident Population by Sex, Age, Race, and Hispanic Origin for the United States and States: April 1, 2020 to July 1, 2021 (PEPASR6H).		

Figure 2.1 shows the geographic distribution of FL24 respondents by zip code centroid. While the proportions of participants in each region generally match Census estimates, it is important to note the clusters of participants in urban areas. This pattern follows the geographic distribution of the U.S. population, but there are relatively few respondents in rural areas, which may complicate some inferences about that segment of the population. Such inferences would likely require oversampling that portion of the population in a subsequent study.

Figure 2.1: Geographic Distribution of FL24 Respondents



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To ensure the protection of survey respondents, the FL24 survey questions and the survey protocol were approved by the Institutional Review Board of the University of Oklahoma. While Verasight was responsible for the recruitment of respondents and implementation of the survey, the University of Oklahoma (IPPRA) manages the data. The survey instrument was programmed to allow the survey to be self-administered at the preferred time and pace of each respondent within defined time constraints to maintain flexibility as well as data quality. Average (median) completion time was 19 minutes. Each respondent who completed the survey received an incentive (survey points) from Verasight. Decisions to participate were entirely voluntary.

3. DATA WEIGHTING

To enhance the demographic representativeness of the sample and generalizability of the results, FL24 employed the data weighting methodology that is described in this section. Weighting survey data to selected demographic characteristics of the target population (also known as sample balancing or post-stratification) provides three key analytical benefits:

1. Generalizability of findings is strengthened to the degree that responses from survey participants are adjusted to mirror the demographic characteristics of the U.S. adult population at the time the survey is administered;
2. Comparability to other studies is strengthened because data weighting minimizes the demographic differences that may arise between this and other samples that match the demographic characteristics of the U.S. adult population;
3. Reliability of trends over time is strengthened because survey data are adjusted to represent continually evolving demographics of the U.S. population, such as the growth

of ethnic and racial minority groups. This is especially important for understanding measures that may be influenced by shifts in national demographic characteristics.

To access these benefits, IPPRA calculated a weight for each FL24 respondent that accounts for imbalances in gender, age, race, and Hispanic ethnicity within each of the four NWS regions that divide the contiguous U.S. (CONUS)—the Eastern, Southern, Central, and Western regions. The weighting process involved three steps:

1. Calculate the proportion of the U.S. population that shares the demographic characteristics of each respondent (the population proportion);
2. Calculate the proportion of the sample that shares the demographic characteristics of each respondent (the sample proportion);
3. Divide the population proportion by the sample proportion to calculate a weight for each respondent.

This process resulted in a weight factor that indicates how much each case will "count" in weighted analyses. A weight factor of one means that responses from a specified participant are used without adjustment. A weight factor greater than one means that a participant with a given set of demographic attributes is underrepresented in the survey sample (relative to the national population), and responses from that participant receive greater statistical emphasis than responses from survey participants who are represented in direct proportion to the adult population. Conversely, a weight factor smaller than one means that a respondent having a given set of demographic attributes is overrepresented in the survey sample (relative to the adult population), and responses from that participant receive less emphasis than fellow respondents who are represented in direct proportion to the adult population. Weight factors were calculated within NWS regions to facilitate generalization within and comparison across the regions.

4. SURVEY INSTRUMENT, WEIGHTED RESPONSE FREQUENCIES, AND CENTRAL TENDENCIES

This section provides a reproduction of the FL24 survey instrument with weighted frequencies or central tendencies for the questions that elicited numeric responses [shown in blue].

You are invited to participate in a research study entitled "Severe Weather and Society." This study seeks to assess how U.S. residents receive, understand, and respond to weather forecasts and warnings. You were selected as a possible participant because you volunteered to participate in online surveys through Verasight or one of its partners. If you agree to participate, you will complete this approximately 20-minute online survey. There are no risks or benefits.

If you participate, you will be compensated according to your agreement with your online survey provider. Your participation is voluntary, and your responses will be de-identified before they are shared for research purposes or published.

Even if you choose to participate now, you may stop participating at any time and for any reason. Your data may be used in future research studies unless you contact me to withdraw your data.

Data are collected via an online survey system that has its own privacy and security policies for keeping your information confidential. The University of Oklahoma cannot provide assurances as to how this online survey system is permitted to use the data you provide.

If you have questions about this research, please contact the Institute for Public Policy Research and Analysis at the University of Oklahoma, at (405) 325-1720 or at ippra@ou.edu.

You can also contact the University of Oklahoma – Norman Campus Institutional Review Board at 405-325-8110 or irb@ou.edu with questions, concerns, or complaints about your rights as a research participant, or if you don't want to talk to the researcher.

By answering the survey questions, I agree to participate in this research. Please print this page for your records. This research has been approved by the University of Oklahoma, Norman Campus IRB.

IRB Number: 9418
Approval date: 10/18/2023

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age: How old are you? [DROPDOWN; Less than 18 to 110; IF < 18 SKIP TO END OF SURVEY] [Median = 50]

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gend: Are you male or female?
0 - Female [50.00%]
1 - Male [50.00%]

hisp: Do you consider yourself to be Hispanic, Latino, or Spanish or to have Hispanic, Latino, or Spanish origins?
0 - No [86.50%]
1 - Yes [13.50%]

race: Which of the following best describes your race?
1 - White [76.83%]
2 - Black or African American [12.17%]
3 - American Indian or Alaska Native [1.50%]
4 - Asian [4.92%]
5 - Native Hawaiian or Pacific Islander [0.17%]
6 - Two or more races [4.42%]

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state: Please select the state or district where your current residence is located.

zip: What is the five-digit zip code at your residence? [VERBATIM; REQUIRE 5-DIGIT NUMERIC]

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How many years have you lived at your current address or any other address within the same zip code area?
long_years: [DROPDOWN; Less than 1 year to 100] [Median = 10]

----------End Web pg----------

[SHOW IF **long_years** < 5]

last_state: Using the dropdown list, please select the state or district where your previous residence was located.

----------End Web pg----------
[SHOW IF **last_state** = Other]

last_state_othr_spec: Other location (please specify) [VERBATIM]

----------End Web pg----------
[SHOW IF **last_state** =/ Other]

last_zip: What was the five-digit zip code where your previous residence was located? [VERBATIM; REQUIRE 5-DIGIT NUMERIC]

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now: Please indicate which of the following statements applies to you.
0 - I am completing this survey from my current primary residence. [89.71%]
1 - I am completing this survey from a location that is not my current primary residence. [10.29%]

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rural: Which of the following categories best describes the location of your current primary residence?
1 - Urban lot in a densely populated area [28.62%]
2 - Suburban lot in a neighborhood that is near a densely populated area [53.68%]
3 - Rural lot in a sparsely populated area [17.70%]

home: Which of the following categories best describes the nature of your current primary residence?
1 - Stand-alone (detached) permanent structure such as a house [65.26%]
2 - Condominium, townhouse, or duplex that is attached to another structure [12.61%]
3 - Apartment or dormitory room that is part of a larger residential complex [16.26%]
4 - Mobile home (whether placed on a permanent foundation or not) [4.98%]
5 - Boat, boathouse, ship, dock, or other floating structure [0.25%]
6 - Other type (please specify) [0.64%]
home_spec: [VERBATIM]

rent: Which of the following categories best describes your living arrangements at your current primary residence?
1 - Live with family or friends (e.g., in someone else's home) and do not pay rent [23.20%]
2 - Pay to rent or lease your primary residence (includes college or other dormitory rooms) [26.59%]
3 - Own your primary residence (includes making mortgage payments or outright ownership with no mortgage payments) [50.21%]

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adults: *Including yourself,* how many *adults* age 18 and older live in your current primary residence? [DROPDOWN; 1 to More than 20] [Median = 2]

children: How many *children* age 17 and younger live in your current primary residence? [DROPDOWN; 0 to More than 20] [Median = 0]

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Now we have some basic questions about the weather. How much do you agree or disagree with the following statements? [RANDOM ORDER]

follow: I follow the weather very closely.
1 - Strongly disagree [1.74%]
2 - Disagree [7.22%]
3 - Neither disagree nor agree [14.98%]
4 – Agree [50.01%]
5 - Strongly agree [26.04%]

plan_around: I plan my daily routine around the weather.
1 - Strongly disagree [3.73%]
2 - Disagree [14.15%]
3 - Neither disagree nor agree [24.30%]
4 - Agree [42.46%]
5 - Strongly agree [15.36%]

und_weather: I don't understand what causes extreme weather events like blizzards, tornadoes, and hurricanes.
1 - Strongly disagree [19.33%]
2 - Disagree [39.57%]
3 - Neither disagree nor agree [20.51%]
4 - Agree [15.73%]
5 - Strongly agree [4.87%]

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How frequently do you get information about the weather from each of the following sources? [RANDOM ORDER IN TABLE]

wthr_info_paper: Newspapers
1 - Never [61.20%]
2 - Less than once per week [10.36%]
3 - About once per week [7.22%]
4 - Several times per week [8.39%]
5 - About once a day [8.15%]
6 - Several times a day [4.67%]

wthr_info_web: Non-government internet websites (such as weather.com)
1 - Never [17.90%]

2 - Less than once per week [17.56%]
3 - About once per week [12.42%]
4 - Several times per week [20.24%]
5 - About once a day [20.18%]
6 - Several times a day [11.71%]

wthr_info_govweb: Government internet websites (such as weather.gov)
1 - Never [39.01%]
2 - Less than once per week [20.72%]
3 - About once per week [8.68%]
4 - Several times per week [12.05%]
5 - About once a day [12.31%]
6 - Several times a day [7.22%]

wthr_info_loctv: Local TV (television) news
1 - Never [17.98%]
2 - Less than once per week [13.89%]
3 - About once per week [9.01%]
4 - Several times per week [17.43%]
5 - About once a day [24.27%]
6 - Several times a day [17.41%]

wthr_info_cabtv: Cable TV (television) news (such as The Weather Channel)
1 - Never [33.34%]
2 - Less than once per week [16.85%]
3 - About once per week [7.52%]
4 - Several times per week [15.12%]
5 - About once a day [15.10%]
6 - Several times a day [12.07%]

wthr_info_radio: Radio
1 - Never [33.01%]
2 - Less than once per week [17.93%]
3 - About once per week [9.92%]
4 - Several times per week [17.31%]
5 - About once a day [12.67%]
6 - Several times a day [9.17%]

wthr_info_fam: Family, friends or colleagues
1 - Never [15.02%]
2 - Less than once per week [21.75%]
3 - About once per week [15.77%]
4 - Several times per week [26.82%]
5 - About once a day [11.63%]
6 - Several times a day [8.99%]

wthr_info_soc: Social Media, such as Facebook
1 - Never [29.63%]
2 - Less than once per week [18.39%]

3 - About once per week [9.67%]
4 - Several times per week [14.44%]
5 - About once a day [13.17%]
6 - Several times a day [14.71%]

wthr_info_app: Weather applications on your phone
1 - Never [6.74%]
2 - Less than once per week [5.06%]
3 - About once per week [6.49%]
4 - Several times per week [17.21%]
5 - About once a day [31.62%]
6 - Several times a day [32.88%]

wthr_info_text: Emergency notifications on your phone
1 - Never [11.21%]
2 - Less than once per week [44.00%]
3 - About once per week [12.05%]
4 - Several times per week [13.24%]
5 - About once a day [9.65%]
6 - Several times a day [9.85%]

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Thinking about all four seasons (spring, summer, fall, winter), how do you rate the risk of the following hazardous weather events to you and the people in the area where you live? [RANDOM ORDER IN TABLE; REQUIRE ALL]

risk_hail: Hail
1 - No risk [7.39%]
2 - Low risk [34.04%]
3 - Moderate risk [36.06%]
4 - High risk [15.19%]
5 - Extreme risk [7.32%]

risk_wind: High winds
1 - No risk [2.08%]
2 - Low risk [13.43%]
3 - Moderate risk [40.34%]
4 - High risk [30.26%]
5 - Extreme risk [13.88%]

risk_lignt: Lightning
1 - No risk [2.95%]
2 - Low risk [20.10%]
3 - Moderate risk [33.71%]
4 - High risk [30.14%]
5 - Extreme risk [13.10%]

risk_heat: Heat waves
1 - No risk [1.30%]

2 - Low risk [9.98%]
3 - Moderate risk [32.29%]
4 - High risk [31.87%]
5 - Extreme risk [24.56%]

risk_drought: Drought
1 - No risk [10.11%]
2 - Low risk [29.33%]
3 - Moderate risk [28.85%]
4 - High risk [19.71%]
5 - Extreme risk [12.00%]

risk_cold: Cold temperatures
1 - No risk [9.25%]
2 - Low risk [24.82%]
3 - Moderate risk [32.76%]
4 - High risk [22.24%]
5 - Extreme risk [10.92%]

risk_snow: Snow
1 - No risk [23.78%]
2 - Low risk [24.18%]
3 - Moderate risk [23.82%]
4 - High risk [17.31%]
5 - Extreme risk [10.92%]

risk_ice: Ice or freezing rain
1 - No risk [17.31%]
2 - Low risk [25.04%]
3 - Moderate risk [27.80%]
4 - High risk [20.72%]
5 - Extreme risk [9.12%]

risk_tor: Tornadoes
1 - No risk [17.37%]
2 - Low risk [28.71%]
3 - Moderate risk [27.43%]
4 - High risk [15.21%]
5 - Extreme risk [11.28%]

risk_flood: Floods
1 - No risk [8.30%]
2 - Low risk [33.39%]
3 - Moderate risk [30.85%]
4 - High risk [16.98%]
5 - Extreme risk [10.48%]

risk_hur: Hurricanes
1 - No risk [42.33%]

2 - Low risk [21.37%]
3 - Moderate risk [13.92%]
4 - High risk [11.58%]
5 - Extreme risk [10.80%]

risk_fire: Wildfires
1 - No risk [17.57%]
2 - Low risk [34.22%]
3 - Moderate risk [21.42%]
4 - High risk [15.85%]
5 - Extreme risk [10.93%]

risk_bliz: Blizzards
1 - No risk [36.52%]
2 - Low risk [22.98%]
3 - Moderate risk [20.05%]
4 - High risk [11.81%]
5 - Extreme risk [8.64%]

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risk_tie: Please indicate which type of event poses the *biggest* risk to you and the people in the area where you live.
1 - Hail [0.98%]
2 - High winds [7.79%]
3 - Lightning [4.77%]
4 - Heat waves [19.65%]
5 - Drought [3.87%]
6 - Cold temperatures [4.92%]
7 - Snow [6.97%]
8 - Ice or freezing rain [5.85%]
9 - Tornadoes [16.65%]
10 - Floods [8.65%]
11 - Hurricanes [11.24%]
12 - Wildfires [5.77%]
13 - Blizzards [2.89%]

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The National Weather Service (NWS), an agency of the United States government, issues weather forecasts and different kinds of alerts to the public about hazardous weather, including severe weather watches, warnings, and advisories.

alert_und: In general, do you understand the difference between watches, warnings, and advisories?
1 - Definitely no [1.13%]
2 - Probably no [4.87%]
3 - Not sure [10.64%]
4 - Probably yes [42.66%]
5 - Definitely yes [40.71%]

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The next few questions focus on floods. For simplicity, we use the word flood to reference all kinds of flooding caused by weather conditions, including flash floods, riverine floods, coastal flooding, and urban flooding.

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To begin, please think about you, family members you live with, and your home as you respond to the following questions about flooding.

fl_risk_prob: How likely do you think it is that the area where you live will be affected by a flood in the next few years?
1 - Not at all likely [13.70%]
2 - Not very likely [36.56%]
3 - Somewhat likely [27.54%]
4 - Very likely [16.03%]
5 - Extremely likely [6.17%]

fl_risk_imp: If the area where you live is affected by a flood in the next few years, how likely do you think it is that you personally will experience negative impacts from the flood?
1 - Not at all likely [12.93%]
2 - Not very likely [39.46%]
3 - Somewhat likely [29.84%]
4 - Very likely [11.71%]
5 - Extremely likely [6.06%]

fl_risk_sev: If the area where you live is affected by a flood in the next few years, how severe do you think the negative impacts will be to you from the flood?
1 - Not at all severe [14.46%]
2 - Not very severe [43.60%]
3 - Somewhat severe [27.21%]
4 - Very severe [9.38%]
5 - Extremely severe [5.36%]

fl_risk_aff: If the area where you live is affected by a flood in the next few years, how worried are you that it will affect you?
1 - Not at all worried [16.42%]
2 - Not very worried [37.15%]
3 - Somewhat worried [29.14%]
4 - Very worried [11.17%]
5 - Extremely worried [6.13%]

----------End Web pg----------

The National Weather Service issues multiple public alerts during floods depending on what type of flooding is occurring. We want to know how you interpret these different types of alerts.

flood_wwa_und: How would you rate your understanding of *flood* WATCHES and WARNINGS?
1 - Poor [2.82%]

2 - Fair [15.05%]
3 - Good [30.78%]
4 - Very good [31.59%]
5 - Excellent [19.75%]

flash_wwa_und: How would you rate your understanding of *flash flood* WATCHES and WARNINGS?
1 - Poor [3.32%]
2 - Fair [15.70%]
3 - Good [29.80%]
4 - Very good [30.33%]
5 - Excellent [20.85%]

----------End Web pg----------

To the best of your knowledge, is the following alert considered a flood WATCH or flood WARNING? [RANDOM SPLIT; 50% get **flood_wwa_watch**; 50% get **flood_wwa_warn**]

flood_wwa_watch: This alert is issued when conditions are favorable for flooding. It does not mean flooding will occur, but it is *possible*.
1 - Flood WATCH [74.98%]
2 - Flood WARNING [21.87%]
3 - Don't know [3.15%]

flood_wwa_warn: This alert is issued when flooding is imminent or occurring.
1 - Flood WATCH [23.25%]
2 - Flood WARNING [72.72%]
3 - Don't know [4.02%]

----------End Web pg----------

flood_warn_time: If the National Weather Service issues a *flood WARNING* for your location, about how much time do you have before flooding begins?
1 - less than 1 hour [24.72%]
2 - 1 to 24 hours [52.94%]
3 - 1 to 2 days [14.78%]
4 - 2 to 4 days [5.52%]
5 - 4 to 8 days [2.04%]

flood_watch_time: If the National Weather Service issues a *flood WATCH* for your location, about how much time do you have before flooding begins?
1 - less than 1 hour [9.49%]
2 - 1 to 24 hours [48.48%]
3 - 1 to 2 days [31.19%]
4 - 2 to 4 days [7.33%]
5 - 4 to 8 days [3.51%]

----------End Web pg----------

flood_flash_und: Do you understand the difference between *floods* and *flash floods*?

1 - Definitely no [2.46%]
2 - Probably no [7.31%]
3 - Not sure [16.94%]
4 - Probably yes [46.64%]
5 - Definitely yes [26.66%]

flood_flash_desc: Please use a few words to explain the primary difference between floods and flash floods. [VERBATIM]

----------End Web pg----------

To the best of your knowledge, is the following alert considered a FLOOD warning or a FLASH FLOOD warning? [RANDOM SPLIT; 50% get **flood_warn**; 50% get **flash_warn**]

flood_warn: This alert is issued for flooding that normally occurs more than six hours after the rainfall begins, and may cover a large area. This results in a gradual ponding or buildup of water in low-lying, flood prone areas, as well as small creeks and streams.
1 - FLOOD warning [68.86%]
2 - FLASH FLOOD warning [25.66%]
3 - Don't know [5.48%]

flash_warn: This alert is issued for flooding that normally occurs within six hours of heavy or intense rainfall. This results in small creeks and streams quickly rising out of their banks.
1 - FLOOD warning [33.97%]
2 - FLASH FLOOD warning [60.85%]
3 - Don't know [5.18%]

----------End Web pg----------

For people who live on the coast of an ocean, coastal flooding and storm surge are also concerns.

coast_live: Do you live on the coast of an ocean?
0 - No [81.75%]
1 - Yes [18.25%]

coast_risk_prob: How likely do you think it is that the area where you live will be affected by *coastal flooding* in the next few years?
1 - Not at all likely [52.49%]
2 - Not very likely [21.92%]
3 - Somewhat likely [13.61%]
4 - Very likely [9.05%]
5 - Extremely likely [2.93%]

surge_risk_prob: How likely do you think it is that the area where you live will be affected by *storm surge* in the next few years?
1 - Not at all likely [36.57%]
2 - Not very likely [24.13%]
3 - Somewhat likely [24.20%]
4 - Very likely [10.82%]

5 - Extremely likely [4.28%]

flood_srg_und: For some people, it is difficult to explain the difference between flooding and storm surge. How would you rate your understanding of the difference between the two hazards?
1 - Poor [18.14%]
2 - Fair [24.64%]
3 - Good [29.65%]
4 - Very good [18.50%]
5 - Excellent [9.07%]

----------End Web pg ----------

The next set of questions is about the variety of forecast information that you might receive when a flood threatens your location. If you have not been in this situation, please think about the information you might receive if you are in this situation in the future.

How much do you agree with the following statements? [RANDOM ORDER]

rec_most_fl: I receive pretty much all of the information that I need for my location.
1 - Strongly disagree [1.06%]
2 - Disagree [4.96%]
3 - Neither disagree nor agree [17.40%]
4 - Agree [55.46%]
5 - Strongly agree [21.13%]

rec_miss_fl: Sometimes I miss information that I need for my location.
1 - Strongly disagree [8.53%]
2 - Disagree [30.60%]
3 - Neither disagree nor agree [28.27%]
4 - Agree [26.77%]
5 - Strongly agree [5.83%]

rec_time: I receive new information about my location as soon as it is available.
1 - Strongly disagree [1.49%]
2 - Disagree [6.31%]
3 - Neither disagree nor agree [18.14%]
4 - Agree [53.09%]
5 - Strongly agree [20.97%]

----------End Web pg ----------

[SHOW IF **rec_most** < 3]

rec_most_exp: It sounds like you are not sure that you receive all the information you need when a flood threatens your location. Can you tell us what information you need but do not receive? [VERBATIM]

----------End Web pg ----------

rec_dif_sit: Can you think of a situation that might cause you to miss important forecast information if a flood were threatening your location? [VERBATIM]

----------End Web pg ----------

flood_hist: Have you ever experienced a flooding event?
0 - No [50.91%]
1 - Yes [49.09%]

----------End Web pg ----------

[SHOW IF **flood_hist** = 1]

Think about the most recent flooding event that you experienced.

flood_long: How long ago was it?
1 - Less than a year ago [11.62%]
2 - 1-2 years ago [17.27%]
3 - 2-5 years ago [21.41%]
4 - More than 5 years ago [49.70%]

flood_type: Do you remember what caused the flooding event?
1 - River flood - flooding caused by a river overflowing its banks [17.26%]
2 - Flash flood - sudden, intense flooding caused by heavy rainfall in a short time period [38.83%]
3 - Coastal flood - flooding caused by ocean water invading land areas along the coast due to high or rising tides [4.96%]
4 - Storm surge - flooding caused by the abnormal rise of ocean water generated by a storm (such as a hurricane or tropical storm) [12.02%]
5 - Urban flood - flooding caused by overwhelmed drainage systems in urban areas [13.62%]
6 - Groundwater flood - flooding caused by saturated groundwater rising to the surface [6.93%]
7 - Dam/levee break flood - flooding caused by a dam or levee failure [1.21%]
8 - Don’t know/don’t remember [5.17%]

flood_act: Did you take some kind of protective action in response to the flooding event?
0 - No [46.54%]
1 - Yes [53.46%]

----------End Web pg ----------

[SHOW IF **flood_act** = 1]

flood_act_kind: What kind of protective action did you take in response to the flooding event? [VERBATIM]

flood_act_satis: Looking back, how satisfied are you with the actions you took?
1 - Very dissatisfied [0.71%]
2 - Dissatisfied [3.07%]
3 - Neither dissatisfied nor satisfied [11.43%]
4 - Satisfied [49.60%]
5 - Very satisfied [35.19%]

flood_act_again: How likely is it that you would take the same action again if you found yourself in the same situation in the future?
1 - Very unlikely [2.02%]
2 - Somewhat unlikely [3.87%]
3 - About as likely as not [10.01%]
4 - Somewhat likely [30.03%]
5 - Very likely [54.08%]

----------End Web pg ----------

[SHOW IF **flood_act_satis** < 3]

flood_act_exp: Can you briefly explain why you were not satisfied with the actions you took? [VERBATIM]

----------End Web pg ----------

People take many different actions in response to flooding events (e.g., look for more information, charge phones, prepare home, evacuate). Think about the actions you take when you get information that a flooding event may impact your location. If you have not been in this situation, please think about the actions you might take if you are in this situation in the future.

How much do you agree with the following statements? [RANDOM ORDER]

resp_ignore_fl: Sometimes I see information that says I could be at risk and decide to take my chances.
1 - Strongly disagree [9.96%]
2 - Disagree [24.07%]
3 - Neither disagree nor agree [25.27%]
4 - Agree [33.40%]
5 - Strongly agree [7.30%]

resp_always_fl: I almost always take the protective actions that officials suggest, even if the actions are difficult (e.g., costly or time consuming).
1 - Strongly disagree [2.07%]
2 - Disagree [12.74%]
3 - Neither disagree nor agree [28.34%]
4 - Agree [42.22%]
5 - Strongly agree [14.63%]

resp_know_fl: Sometimes I am not sure what to do or what protective actions to take.
1 - Strongly disagree [10.03%]
2 - Disagree [28.06%]
3 - Neither disagree nor agree [27.83%]
4 - Agree [28.77%]
5 - Strongly agree [5.31%]

----------End Web pg ----------

resp_dif_sit: Can you think of a situation that might cause you to not take protective action if a flooding event were threatening your location? [VERBATIM]

----------End Web pg----------

[RANDOMIZE; 50% **home_ff_do** | 50% **car_ff_do**]

home_ff_do: Imagine that it is Saturday night at 8:00 PM and you are at home watching a movie. Suddenly, you see a FLASH FLOOD WARNING indicating that there is a high risk for flash flooding in the next 60 minutes. What would you do? Please be as specific as possible. [VERBATIM]

----------End Web pg----------

car_ff_do: Imagine that it is Saturday night at 8:00 PM and you are driving to the grocery store. Suddenly, you see a FLASH FLOOD WARNING indicating that there is a high risk for flash flooding in the next 60 minutes. What would you do? Please be as specific as possible. [VERBATIM]

----------End Web pg----------

ff_do_conff: How confident are you that your actions would protect you from the flash flood?
1 - Not at all confident [2.42%]
2 - Not very confident [9.30%]
3 - Somewhat confident [40.82%]
4 - Very confident [33.71%]
5 - Extremely confident [13.76%]

----------End Web pg----------

When flooding events threaten your area, how much do you rely on the following sources of information? If you have not been in this situation, how much would you plan to rely on the following sources of information if you were traveling in an area that is threatened by a flooding event? [RANDOM ORDER IN TABLE]

rely_nws: National Weather Service
1 - Not at all [1.57%]
2 - Not very much [4.33%]
3 - Somewhat [23.13%]
4 - Very much [45.31%]
5 - Extensively [25.67%]

rely_loctv: Local TV stations
1 - Not at all [7.33%]
2 - Not very much [7.48%]
3 - Somewhat [24.42%]
4 - Very much [37.16%]
5 - Extensively [23.62%]

rely_fam: Family, friends, neighbors, employers, co-workers, etc.
1 - Not at all [6.14%]
2 - Not very much [15.36%]

3 - Somewhat [38.19%]
4 - Very much [29.73%]
5 - Extensively [10.58%]

rely_wx_chan: National TV stations like The Weather Channel or WeatherNation
1 - Not at all [6.08%]
2 - Not very much [11.09%]
3 - Somewhat [28.60%]
4 - Very much [36.55%]
5 - Extensively [17.68%]

rely_em: State or local emergency managers
1 - Not at all [3.18%]
2 - Not very much [6.94%]
3 - Somewhat [26.59%]
4 - Very much [39.50%]
5 - Extensively [23.79%]

----------End Web pg----------

[SHOW IF **rely_nws** > 2]

You indicated that you rely on the National Weather Service for information about flooding events. How do you usually get information from the National Weather Service? Please select all that apply. [RANDOM ORDER CHECK BOXES; 1 = SELECTED]
nws_twit: X (formerly known as Twitter) [16.08%]
nws_face: Facebook [26.51%]
nws_web: Internet website [42.34%]
nws_tv: Television [48.74%]
nws_rad: Weather radio (NOAA/NWS radio) [26.17%]
nws_phone: Cell phone applications or notifications [70.38%]
nws_othr: Other (please specify) [2.40%]
nws_othr_spec: [VERBATIM]

----------End Web pg----------

[SHOW IF **rely_nws** = 1 OR 2]

rely_nws_why: You indicated that you do not rely on the National Weather Service for information about flooding events. Can you tell us why? [RANDOM ORDER]
1 - I don't know where to find the information [34.69%]
2 - The information is usually too complicated for me to understand [10.71%]
3 - They don't provide the information I want or need [18.45%]
4 - I don't trust the National Weather Service [12.30%]
5 - Other (please specify) [23.85%]
rely_nws_why_spec: [VERBATIM]

----------End Web pg----------

When flooding events threaten your area, how much do you rely on the following **channels** of information? Here, we define **channels** as tools or avenues of information. If you have not been in this situation, how much would you plan to rely on the following **channels** of information if you were traveling in an area that is threatened by a flooding event? [RANDOM ORDER IN TABLE]

rely_bdrad: Broadcast radio
1 - Not at all [18.93%]
2 - Not very much [15.75%]
3 - Somewhat [31.35%]
4 - Very much [22.36%]
5 - Extensively [11.61%]

rely_wxrad: Weather radio (NOAA/NWS radio)
1 - Not at all [25.29%]
2 - Not very much [16.54%]
3 - Somewhat [22.27%]
4 - Very much [23.41%]
5 - Extensively [12.48%]

rely_tv: Television
1 - Not at all [11.36%]
2 - Not very much [8.73%]
3 - Somewhat [23.80%]
4 - Very much [33.82%]
5 - Extensively [22.29%]

rely_int: Internet websites focused on weather forecasts, such as those provided by the National Weather Service
1 - Not at all [5.33%]
2 - Not very much [7.13%]
3 - Somewhat [28.86%]
4 - Very much [37.47%]
5 - Extensively [21.22%]

rely_soc: Social media, such as Facebook
1 - Not at all [20.36%]
2 - Not very much [18.50%]
3 - Somewhat [30.80%]
4 - Very much [20.24%]
5 - Extensively [10.10%]

rely_wom: Word-of-mouth (including telephone calls or texts) from family, friends, neighbors, employers, co-workers, etc.
1 - Not at all [7.85%]
2 - Not very much [16.77%]
3 - Somewhat [41.39%]
4 - Very much [25.53%]
5 - Extensively [8.45%]

rely_phone: Automated text or phone notifications

1 - Not at all [4.92%]
2 - Not very much [7.29%]
3 - Somewhat [26.98%]
4 - Very much [38.03%]
5 - Extensively [22.78%]

----------End Web pg ----------

rely_tie: Please indicate which channel you rely on the most for information about flooding events.
1 - Broadcast radio [4.87%]
2 - Weather radio (NOAA/NWS radio) [6.28%]
3 - Television [27.38%]
4 - Internet websites focused on weather forecasts, such as those provided by the National Weather Service [26.45%]
5 - Social media, such as Facebook [7.94%]
6 - Word-of-mouth (including telephone calls or texts) from family, friends, neighbors, employers, co-workers, etc. [4.45%]
7 - Automated text or phone notifications [22.62%]

----------End Web pg ----------

Forecasters, websites, and phone applications often use maps to display information about flood risks.

flood_map_und: How would you rate your understanding of maps?
1 – Poor [5.17%]
2 – Fair [14.25%]
3 – Good [27.64%]
4 - Very good [27.89%]
5 – Excellent [25.05%]

flood_map_pref: Do you think maps are useful for displaying information about flood risks?
0 – No [14.95%]
1 – Yes [85.05%]

----------End Web pg----------

In addition to maps, forecasters often use probabilities to display information about flood risks.

For some people, probabilities can be difficult to calculate and interpret when you are trying to sort through a lot of information at once.

your_ability: How would you rate your ability to calculate and interpret probabilities when making decisions?
1 – Poor [6.08%]
2 – Fair [16.55%]
3 – Good [32.95%]
4 - Very good [30.10%]
5 – Excellent [14.32%]

----------End Web pg----------

[RANDOMIZE: **rand_timeline**: flooding event | flash flooding event]

Some people look for different kinds of information at different points in time. In the next few questions, we are going to give you a timeline and ask you to indicate the type of information that is *most* important at each point in time.

five_days: **Five days** before the [**rand_timeline**: flooding event | flash flooding event]
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

%	flooding event	flash flooding event
Location	32.75	38.99
Timing	11.50	11.85
Chance	28.41	23.81
Severity	8.97	8.28
Impacts	10.25	10.32
Protective actions	8.12	6.75

three_days: **Three days** before the [**rand_timeline**: flooding event | flash flooding event]
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

%	flooding event	flash flooding event
Location	25.49	23.34
Timing	20.76	21.13
Chance	22.42	20.35
Severity	14.42	18.93
Impacts	11.51	10.48
Protective actions	5.40	5.77

two_days: **Two days** before the [**rand_timeline**: flooding event | flash flooding event]
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

%	flooding event	flash flooding event
Location	14.66	17.50
Timing	20.11	15.92
Chance	21.08	21.97

Severity	23.98	19.01
Impacts	11.23	15.54
Protective actions	8.94	10.06

one_day: **One day** before the [**rand_timeline**: flooding event | flash flooding event]
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

%	flooding event	flash flooding event
Location	16.34	16.78
Timing	16.88	16.27
Chance	7.78	10.92
Severity	20.87	18.06
Impacts	16.15	15.43
Protective actions	21.98	22.53

zero_day: **The day of** the [**rand_timeline**: flooding event | flash flooding event]
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

%	flooding event	flash flooding event
Location	13.55	12.78
Timing	13.73	14.12
Chance	8.47	6.19
Severity	11.74	15.76
Impacts	19.31	15.47
Protective actions	33.20	35.68

----------End Web pg----------

The NWS Weather Prediction Center (WPC) issues Excessive Rainfall Outlooks that indicate flash flooding threats across the contiguous United States. These Outlooks use a combination of words, colors, and numeric probabilities to represent threats. Here is one example:

[SHOW EXAMPLE OF ERO]

ero_see: Have you ever seen an Excessive Rainfall Outlook before?
0 – Yes [25.49%]
1 – No [57.00%]
2 - Not sure [17.50%]

ero_int_sub: How confident are you in your ability to understand the information presented in this Outlook?

1 - Not at all confident [5.92%]
2 - Slightly confident [13.29%]
3 - Moderately confident [32.22%]
4 - Very confident [32.85%]
5 - Extremely confident [15.72%]

ero_int_obj: Based on this Outlook, would you say that **location A** is...
1 - Very *unlikely* to experience flash flooding [3.98%]
2 - Somewhat *unlikely* to experience flash flooding [10.05%]
3 - Somewhat *likely* to experience flash flooding [51.15%]
4 - Very *likely* to experience flash flooding [31.63%]
5 - Not sure [3.19%]

ero_use: How likely are you to use an Outlook like this for making future decisions?
1 - Very unlikely [4.20%]
2 - Somewhat unlikely [8.02%]
3 – Unsure [25.66%]
4 - Somewhat likely [42.47%]
5 - Very likely [19.65%]

----------End Web pg ----------

[If **ero_int_sub** < 3]

[SHOW EXAMPLE OF ERO]

ero_int_no_why: You indicated that you are not confident in your ability to understand this Outlook. Can you tell us what aspects of the graphic are most confusing? [verbatim]

----------End Web pg ----------

[If **ero_use** < 3]

[SHOW EXAMPLE OF ERO]

You indicated that you are unlikely or not very likely to use an Outlook like this for making future decisions. Can you tell us why? Please select all that apply.
ero_use_no_fnd: I don't know where to find forecasts like this. [33.93%]
ero_use_no_inacc: I believe forecasts like this are too inaccurate or unreliable. [30.78%]
ero_use_no_rel: Forecasts like this are not relevant to my daily life or activities. [30.43%]
ero_use_no_int: I find it hard to interpret forecasts like this. [19.76%]
ero_use_no_oth: Other (please specify) [12.01%]
ero_use_oth_spec: [verbatim]

----------End Web pg ----------

Now we are going to show you an example message from the National Weather Service and ask you how you might respond if you got this message. We know this can be difficult without more information. Just give us your first impression.

Imagine that it is Saturday evening at 6:00 PM and you receive the following message on your phone:

wea_rand_img: [RANDOMIZE: warning, emergency, warning_base, warning_cons, warning_cat]

If you saw this message and you were in the impacted area…

ibw_cncrn: How concerned would you be about the flash flood?
1 - Not at all concerned
2 - Not very concerned
3 - Somewhat concerned
4 - Very concerned
5 - Extremely concerned

%	warning	emergency	warning_base	warning_cons	warning_cat
Not at all concerned	4.04	0.61	2.52	1.12	1.49
Not very concerned	3.01	3.31	6.42	5.92	4.25
Somewhat concerned	26.27	31.48	40.55	38.73	30.52
Very concerned	40.98	39.97	29.87	34.06	32.32
Extremely concerned	25.70	24.63	20.63	20.53	31.42

ibw_act: Would you take actions to protect yourself from the flash flood?
1 - Definitely no
2 - Probably no
3 - Not sure
4 - Probably yes
5 - Definitely yes

%	warning	emergency	warning_base	warning_cons	warning_cat
Definitely no	1.75	0.72	0.52	1.06	1.09
Probably no	6.99	4.29	11.30	10.62	6.92
Not sure	8.61	12.66	15.34	14.61	14.67
Probably yes	46.17	43.28	38.06	41.45	35.07
Definitely yes	36.47	39.06	34.78	32.26	42.25

ibw_act_conf: How confident are you that the actions you might take to protect yourself from the flash flood would be effective?
1 - Not at all confident
2 - Not very confident
3 - Somewhat confident
4 - Very confident
5 - Extremely confident

%	warning	emergency	warning_base	warning_cons	warning_cat
Not at all confident	2.52	0.35	0.16	1.54	1.50
Not very confident	7.76	6.64	8.92	11.12	6.10
Somewhat confident	32.78	38.78	31.56	33.67	37.52
Very confident	39.58	39.74	35.43	40.40	37.28
Extremely confident	17.36	15.49	23.93	13.28	17.59

----------End Web pg ----------

In many places, homeowners insurance policies do not cover damage due to flooding and separate flood insurance policies must be purchased to cover that type of damage.

Do you currently have any of the following types of insurance on your home? [Please indicate all that apply] [RANDOMIZED ORDER EXCEPT **ins_none** ALWAYS LAST]
ins_home: Standard homeowners insurance [65.36%]
ins_flood: Flood insurance [18.82%]
ins_none: None of the above/Not applicable [28.85%]

----------End Web pg----------

[SHOW ONLY IF **ins_home** = 1 AND **ins_flood** = 0]

home_cover: Does your homeowners insurance cover damage from flooding?
0 – No [35.52%]
1 – Yes [24.60%]
2 - Not sure [39.88%]

----------End Web pg ----------

[SHOW ONLY IF **ins_flood** = 0]

ins_flood_ever: Have you ever had flood insurance, separate from homeowners insurance, on your home?
0 – No [77.37%]
1 – Yes [9.99%]
2 - Not sure [12.64%]

----------End Web pg ----------

[SHOW ONLY IF **ins_flood** = 1]

flood_mortgage_ins: Are you required to have flood insurance as part of a mortgage (or escrow) on your home?
0 – No [45.01%]
1 – Yes [47.14%]
2 - Not sure [7.85%]

----------End Web pg ----------

As you may know, the Federal Emergency Management Agency (FEMA) is responsible for developing and maintaining a map of areas that are susceptible to flooding in the U.S. These areas are called Special Flood Hazard Areas or flood zones, or floodplains for short.

fema_flood_zone: To the best of your knowledge, is your home located in a FEMA flood zone?
0 – No [62.01%]
1 – Yes [11.97%]
2 - Not sure [26.02%]

----------End Web pg ----------

flood_damage: Has your home ever been damaged by a flooding event?
0 – No [82.27%]
1 – Yes [13.14%]
2 - Not sure [4.58%]

----------End Web pg ----------

[SHOW ONLY IF **flood_damage** = 1]

flood_damage_yrs: Approximately how long ago did damage caused by the most recent flooding event occur?
1 - 0 - 5 years ago [21.97%]
2 - 6 -10 years ago [34.29%]
3 - 11-15 years ago [22.17%]
4 - 16 - 20 years ago [10.55%]
5 - 21 - 25 years ago [3.48%]
6 - More than 25 years ago [7.53%]

----------End Web pg ----------

flood_prob_1yr: If you had to guess, what is the probability that your home will be damaged by a flooding event next year?
1 - 0% (no chance) [27.11%]
2 - 1-10% (very low chance) [43.84%]
3 - 11-25% (low chance) [18.29%]
4 - 26-50% (medium chance) [8.83%]
5- 51-75% (high chance) [1.25%]
6 - 76-100% - (very high chance) [0.68%]

flood_prob_30yr: What is the probability that your home will be damaged by a flooding event in the next 30 years?
1 - 0% (no chance) [18.94%]
2 - 1-10% (very low chance) [39.21%]
3 - 11-25% (low chance) [21.82%]
4 - 26-50% (medium chance) [14.52%]
5- 51-75% (high chance) [4.29%]
6 - 76-100% - (very high chance) [1.22%]

----------End Web pg ----------

income: Was the estimated annual income for your household in 2023:
1 - Less than $50,000 [go to inc50] [24.89%]
2 - At least $50,000 but less than $100,000 [go to inc100] [40.24%]
3 - At least $100,000 but less than $150,000 [go to inc150] [21.47%]
4 - $150,000 or more [go to inc200] [13.40%]

----------End Web pg ----------

inc_50: Was the estimated annual income for your household in 2023:
1 - Less than $10,000 [27.51%]
2 - $10,000 to less than $20,000 [17.56%]
3 - $20,000 to less than $30,000 [19.89%]
4 - $30,000 to less than $40,000 [17.10%]
5 - $40,000 to less than $50,000 [17.94%]

----------End Web pg ----------

inc_100: Was the estimated annual income for your household in 2023:
6 - $50,000 to less than $60,000 [27.18%]
7 - $60,000 to less than $70,000 [22.70%]
8 - $70,000 to less than $80,000 [23.00%]
9 - $80,000 to less than $90,000 [11.19%]
10 - $90,000 to less than $100,000 [15.93%]

----------End Web pg ----------

inc_150: Was the estimated annual income for your household in 2023:
11 - $100,000 to less than $110,000 [21.66%]
12 - $110,000 to less than $120,000 [18.57%]
13 - $120,000 to less than $130,000 [15.14%]
14 - $130,000 to less than $140,000 [18.61%]
15 - $140,000 to less than $150,000 [26.01%]

----------End Web pg ----------

inc_200: Was the estimated annual income for your household in 2023:
16 - $150,000 to less than $160,000 [13.32%]
17 - $160,000 to less than $170,000 [6.91%]
18 - $170,000 to less than $180,000 [10.35%]
19 - $180,000 to less than $190,000 [10.43%]
20 - $190,000 to less than $200,000 [10.40%]
21 - $200,000 or more [48.58%]

----------End Web pg ----------

edu: What is the highest level of education you have COMPLETED?
1 - Less than high school [1.90%]
2 - High school / GED [22.18%]
3 - Vocational or Technical Training [6.14%]
4 - Some College; NO degree [16.13%]
5 - 2-year College / Associate's degree [8.33%]
6 - Bachelor's Degree [28.32%]
7 - Master's Degree [14.01%]
8 - PhD / JD (Law) / MD [3.00%]

----------End Web pg ----------

comments: Is there anything else that you would like us to know about how you receive, understand, or respond to information about flooding?